

Analyzing English-Language Texts about Korea: Exploratory Text Analytics Final Project

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Abstract—This project constructs a digital critical edition of a corpus of four English-language books about Korea written between 1882 and 1908 by Western authors with distinct national backgrounds and stances. This project uses TF-IDF vectorization, Principal Component Analysis (PCA), Latent Dirichlet Allocation (LDA), and Word2Vec word embeddings to uncover cultural patterns embedded in how Westerners wrote about Korea in English language during a period of intense geopolitical competition over the Korean peninsula. The results converge on a consistent picture: most authors imposed foreign cultural, political, and religious frameworks onto their accounts of Korea. Homer Hulbert emerges as a consistent outlier across all four methods. His linguistic patterns reflect his deep immersion in Korean culture and signal his later role as a Korean independence advocate.

Index Terms—corpus, tokenization, vectorization, exploratory text analytics, TF-IDF, PCA, LDA, Word2Vec, Korea

I. INTRODUCTION

This project constructs a digital critical edition of a corpus of English-language books about Korea drawn from Project Gutenberg. The four texts were written by Western authors between 1882 and 1908, during the period of intense geopolitical competition over the Korean Peninsula that involved Japan, China, Russia and the United States. By applying computational text analysis methods to these texts, this project aims to uncover cultural patterns embedded in how outsiders wrote about Korea during this period.

The study is guided by three research questions: (1) What vocabulary is most distinctive to each author and what does it reveal about their cultural frame? (2) Do unsupervised algorithms show ideological or cultural axes consistent with the authors’ known stances? (3) How are key countries outside Korea semantically constructed across the corpus?

II. CORPUS OVERVIEW

The corpus consists of four texts sourced from Project Gutenberg. Table I summarizes the key metadata for each book. The four authors represent meaningfully different national backgrounds, motivations, and stances toward Korea.

TABLE I: Corpus Metadata (LIBRARY Table)

Author	Title	Year	Nationality
Griffis	<i>Corea: The Hermit Nation</i>	1882	American
Bishop	<i>Korea and Her Neighbours</i>	1897	British
Hulbert	<i>The Passing of Korea</i>	1905	American
Ladd	<i>In Korea with Marquis Ito</i>	1908	American

Griffis wrote as a historian relying on secondary sources and missionary accounts, never having visited Korea. Bishop was a travel writer who visited Korea multiple times and grew increasingly sympathetic to the Korean people. Hulbert was an educator who lived in Korea for years, learned the Korean language, and eventually became a Korean independence advocate. Ladd was a philosopher who visited Korea under Japanese auspices and produced a work broadly supportive of Japanese colonial administration.

III. METHODS

A. Corpus Construction and Tokenization (F0–F3)

Source texts were retrieved from Project Gutenberg using the `requests` library. Each text was parsed into a hierarchical OHCO (Ordered Hierarchy of Content Objects) structure (book, chapter, paragraph, sentence, token) and stored as a TOKEN table. Each token was annotated with part-of-speech tags, stems, and stopword flags using NLTK. Chapter segmentation required custom regular expressions for each book as only Bishop follows a standard chapter-heading pattern.

B. TF-IDF Vectorization (F4)

The TOKEN table was aggregated into a Bag-of-Words (BOW) matrix, from which term frequency (TF) and inverse document frequency (IDF) were computed. The resulting TF-IDF matrix assigns high scores to terms distinctive to a given book and low scores to terms shared across the corpus. The VOCAB table records the IDF and aggregate TF-IDF weight for each term.

C. Unsupervised Methods (F5)

Three unsupervised methods were applied to the vectorized corpus.

PCA was performed via eigendecomposition of a term covariance matrix, projecting the four books into a low-dimensional document-component space (DCM). Component loadings were inspected to aid interpretation.

LDA was applied to a raw-count document-term matrix (5,000 features, `min_df=1`) to identify five latent topics. The model produces a document-topic matrix (THETA) and a topic-term matrix (PHI).

Word2Vec was trained on 26,789 sentence-level token sequences (vector size = 100, window = 5, `min_count` = 5, epochs = 10, seed = 42). Nearest-neighbor probes on culturally

salient terms were used to interpret the resulting semantic space.

IV. RESULTS

A. POS Distribution

Part-of-speech analysis revealed meaningful differences in authorial style. Hulbert’s book contains more verbs than the other three, consistent with his historical narrative style in which actions and events are foregrounded. Bishop’s book contains the most adjectives, consistent with her travel-writing genre, in which descriptive and sensory language is expected.

TABLE II: POS Distribution (Proportion of Alphabetic Tokens)

Author	ADJ	ADV	NOUN	OTHER	VERB
Hulbert	0.06	0.04	0.29	0.44	0.17
Griffis	0.08	0.04	0.30	0.43	0.15
Bishop	0.09	0.04	0.29	0.43	0.15
Ladd	0.09	0.05	0.27	0.45	0.15

B. TF-IDF: Distinctive Vocabulary by Author

Tab. III shows the top 10 TF-IDF terms for each author. TF-IDF surfaces the most distinctive — rather than the most frequent — vocabulary for each book, making it well-suited for comparing authorial voice and thematic focus.

TABLE III: Top 10 TF-IDF Terms per Author

Griffis		Bishop	
Corean	0.00435	dæmon	0.00103
Corea	0.00276	Phyöng	0.00046
Chō	0.00096	Wön	0.00035
sen	0.00088	Wladivostok	0.00023
mikado	0.00049	dōng	0.00019
Shinra	0.00047	Chhōn	0.00018
Korai	0.00038	Russian	0.00016
chiu	0.00034	exorcism	0.00016
Manchiu	0.00026	Miller	0.00014
Hiaksai	0.00026	ren	0.00013

Hulbert		Ladd	
Koryū	0.00243	Ito	0.00058
Sil	0.00234	Christian	0.00020
gu	0.00185	Pyeng	0.00018
ryū	0.00177	Russia	0.00014
Pāk	0.00121	jinrikisha	0.00012
yūng	0.00100	redemption	0.00010
Ch	0.00090	Russian	0.00010
P	0.00063	Cabinet	0.00009
wūn	0.00062	Suwon	0.00008
je	0.00059	Hague	0.00008

Hulbert’s top terms are predominantly Korean words in romanized form (*Koryū*, *gu*, *ryū*, *Pāk*), demonstrating deep immersion in Korean culture that sets him apart from all other authors. Griffis relies heavily on ancient kingdom names (*Shinra*, *Korai*) and the archaic exonym *Corean*, reflecting his outsider-historian perspective drawn from secondary sources. Ladd’s distinctive terms include *Ito* — Prince Itō Hirobumi,

the Japanese Resident-General — and geopolitical vocabulary (*Russia*, *Hague*, *Cabinet*), consistent with his pro-Japanese political frame.

C. PCA: Ideological Axes in the Corpus

Fig. 1 shows the four books projected onto their first two principal components. PC0 captures the most variance in the corpus and is interpretable as a **Western-lens vs. Korean-lens** axis. Positive loadings include Western-centric terms (*mikado*, *Russian*, *church*), while negative loadings include romanized Korean vocabulary (*gu*, *wun*, *gyung*). This axis operationalizes a fundamental divide in how the corpus frames Korea: through foreign cultural and political concepts, or through Korean concepts on their own terms.

PC0 vs PC1 (colored by stance)

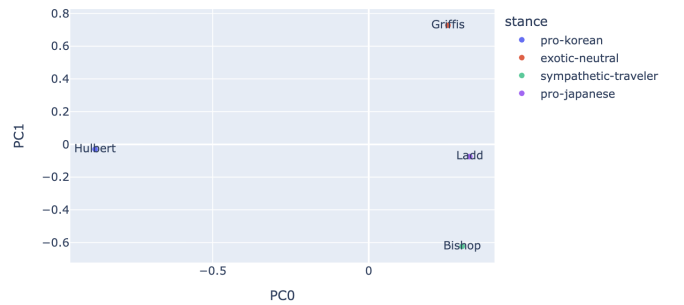


Fig. 1: PCA scatter plot: books projected onto PC0 (x-axis) vs. PC1 (y-axis), colored by authorial stance.

Hulbert is a clear outlier, scoring strongly negative on PC0 — the most Korean-centric position in the corpus — despite being American. This is consistent with his biography: he lived in Korea as an educator, learned the Korean language, championed Korean independence, and was eventually buried in Korea at his own request. Ladd scores highest on PC0, confirming that his text is the most Western-centric. The PCA result suggests that pro-Japanese and Western-centric framing are, in this corpus, linguistically correlated.

D. LDA: Hidden Topics Across the Corpus

TABLE IV: Document-Topic Matrix

Author	Topic 1	Topic 2	Topic 3	Topic 4	Topic 5
Griffis	0.001121	0.001785	0.000002	0.000002	0.997089
Bishop	0.997824	0.002116	0.000003	0.000003	0.000054
Hulbert	0.000003	0.000003	0.000003	0.999989	0.000003
Ladd	0.000141	0.999850	0.000003	0.000003	0.000003

TABLE V: Top Terms per Topic (in order of frequency rank)

Rank	Topic 0	Topic 1	Topic 2	Topic 3	Topic 4
1	korean	korea	marquis	king	japanese
2	one	korean	sincerity	sent	corean
3	korea	japanese	prejudice	one	one
4	japanese	japan	circles	people	corea
5	two	one	descend	japanese	chinese
6	men	government	sections	emperor	japan
7	great	foreign	halted	koryū	china
8	seoul	upon	chon	la	two
9	much	time	community	sil	made
10	chinese	would	hamlets	made	people

Ladd concentrates heavily in Topics 0 and 2 (governance and military), consistent with his focus on Japanese administration. Bishop weights most toward Topic 1 (travel), consistent with her genre. Hulbert loads most on Topic 3 (history), reflecting his treatment of Korean civilization. Griffiths spreads across Topics 0 and 3, consistent with his role as a historian covering both ancient kingdoms and contemporary governance.

E. Word2Vec: Semantic Neighborhoods of Key Terms

Table VI summarizes nearest-neighbor probes on key terms.

TABLE VI: Word2Vec Nearest Neighbors for Selected Terms

Term	Nearest neighbors (top 5)
korea	russia, corea, policy, relations, peace
japan	russia, corea, peace, treaty, china
chinese	japanese, army, invasion, forces, force
america	india, paramount, vital, neutrality, steadily
church	missions, methodist, catholic, roman, protestant

The word *Korea* clusters with *Russia*, *policy*, *relations*, and *peace*, suggesting that Korea was framed primarily as a geopolitical object. *Japan* appears near *Corea*, *peace*, *treaty*, and *China*, reflecting the great-power rivalries that defined Northeast Asia during this period.

The semantic neighborhood of *Chinese* is notably martial: nearest neighbors include *army*, *invasion*, and *forces*, likely anchored to the Sino-Japanese War of 1894–95, which looms large across all four texts. *America*, by contrast, clusters with more abstract terms (*paramount*, *vital*, *neutrality*), suggesting that American presence was discussed in terms of strategic posture rather than direct military engagement.

One notable exception is *church*, which clusters cleanly with *missions*, *Methodist*, *Catholic*, and *Protestant*. The religious register is self-contained and largely insulated from the political vocabulary that dominates the rest of the corpus, suggesting that missionary discourse operated in a separate sphere from the geopolitical narrative.

V. CONCLUSIONS

This project applied four unsupervised methods — TF-IDF, PCA, LDA, and Word2Vec — to a corpus of four English-language books about Korea written in the late 19th to early 20th century.

Colonial-era Western writers constructed Korea primarily through their own cultural and political lenses. PCA and TF-IDF both show the same pattern. Most authors wrote about Korea from their foreign frameworks, including Japanese imperialism, Christian mission, Western diplomatic norms. Korea itself appears as a stage on which foreign powers, armies, and missionaries took their purposes-driven actions on. This is most visible in the Word2Vec results, where the semantic neighborhoods of all major geopolitical actors are defined by power and force, and Korea clusters primarily with terms of negotiation and policy.

Hulbert is consistently an exception from these cultural patterns observed among most authors across every method. For example, his TF-IDF vocabulary is dominated by romanized Korean terms. His PCA score is the most Korean-centric. His LDA result concentrates in the historical topic. And his biography that the author researched following the exploratory text analytics study explains these patterns. Most importantly, no prior biographical knowledge was required for any method to surface his distinctiveness. The linguistic findings from these supervised methods was sufficient.

This project demonstrates that exploratory text analysis can identify hidden cultural stances using machine learning tools quantitatively that literary and historical scholars have long made qualitatively. And it can do so at the level of the individual vocabulary, the paragraph, the chapter, the document, and the corpus.

VI. DECLARATION ON GENERATIVE AI

During the preparation of this work, the author used Claude (Anthropic) and ChatGPT to: create a LaTeX draft using the provided template, and to provide help with code. The author reviewed and edited all content as needed and takes full responsibility.

REFERENCES

- [1] Project Gutenberg, “Free eBooks,” [Online]. Available: <https://www.gutenberg.org>
- [2] W. E. Griffiths, *Corea: The Hermit Nation*. New York: Scribner, 1882.
- [3] I. L. Bishop, *Korea and Her Neighbours*. London: Murray, 1897.
- [4] H. B. Hulbert, *The History of Korea*. Seoul: Methodist Publishing House, 1905.
- [5] G. T. Ladd, *In Korea with Marquis Ito*. New York: Scribner, 1908.
- [6] S. Bird, E. Klein, and E. Loper, *Natural Language Processing with Python*. O’Reilly Media, 2009. [Online]. Available: <https://www.nltk.org>
- [7] E. W. Poitras, “Hulbert, Homer Bezaleel (1863–1949),” *History of Missiology*, Boston University School of Theology, last updated May 21, 2013. [Online]. Available: <https://www.bu.edu/missiology/missionary-biography/g-h/hulbert-homer-bezaleel-1863-1949/>